

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2024

Course: B. Tech.

Semester : V

Branch : Electronics and Telecommunication Engineering/Electronics Engineering

Subject Code & Name: Digital Signal Processing BTETC502_BTETXC502

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Why is quantization necessary in DSP?	CO1	1
a) To improve resolution		
b) To convert to binary		
c) To reduce noise		
d) To amplify signals		
2 Which of the following is NOT a requirement for DSP?	CO1	1
a) Data Sampling		
b) Quantization		
c) Signal Compression		
d) Modulation		
3 The DTFT of $\delta[n]$ (unit impulse) is:	CO2	1
a) 0		
b) 1		
c) $e^{j\omega}$		
d) 2π		

4 Parseval's theorem in DTFT relates:

CO2 1

- a) Energy in time and frequency
- b) Power in time and frequency
- c) Periodicity of signals
- d) Linearity of signals

5 Scaling in the time domain corresponds to _____ in the frequency domain?

CO2 1

- a) Convolution
- b) Compression
- c) Expansion
- d) Time Scaling

6 Which method can be used to compute linear convolution using circular convolution?

CO2 1

- a) Zero-padding
- b) Truncation
- c) Windowing
- d) Scaling

7 The main advantage of FFT over DFT is:

CO4 1

- a) Higher accuracy
- b) Reduced computational complexity
- c) Reduced memory requirements
- d) Simplicity

8 The Z-transform is particularly useful for analyzing:

CO3 1

- a) Continuous systems
- b) Discrete systems
- c) Analog systems
- d) Random systems

9	For Z transform, the ROC must:	CO3	1
	a) Contain infinity b) Contain the unit circle c) Be bounded d) Exclude all poles		
10	Bilinear transformation introduces:	CO5	1
	a) Aliasing b) Frequency warping c) Phase distortion d) Oscillations		
11	The inverse Z-transform is used to:	CO3	1
	a) Find the Z-domain representation b) Convert a signal back to the time domain c) Compute the Fourier transform d) Find the region of convergence		
12	In multirate signal processing, an upsampler is used to:	CO5	1
	a) Increase the sampling rate b) Decrease the sampling rate c) Increase the frequency of the signal d) Decrease the frequency of the signal		
Q. 2 Solve the following.			
	A) Write the applications of Digital Signal Processing.	CO1	6
	B) Explain the term aliasing with neat diagrams. Explain various cases: under-sampling, over-sampling and normal sampling. What can be done to combat the effect of aliasing?	CO1	6
Q.3 Solve the following.			
	A) State and prove any four properties of DTFT.	CO2	6
	B) Determine the DTFT of the signals:	CO2	6
	i) $x(n) = \sin(w_0 n)u(n)$ ii) $x(n) = \cos(w_0 n)u(n)$		

Q. 4 Solve Any Two of the following.**12**

- A) i) Compute the 4-point DFT of $x(n) = \{1, -1, 1, -1\}$ for $0 \leq n \leq 3$
ii) Compute the 4-point IDFT of $X(k) = \{0, -2j, 0, 2j\}$ for $0 \leq k \leq 3$
- B) Use overlap –save method to convolve the following sequences:
 $x(n) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ for $0 \leq n \leq 9$
and $h(n) = \{1, 0, -1\}$ for $0 \leq n \leq 2$
- C) Draw the sample ROC corresponding to the following system characteristics:
i) Causal and stable
ii) Causal and unstable
iii) Anti-causal and stable
iv) Anti-causal and unstable
v) Non-causal and stable
vi) Non-causal and unstable

CO2 6

CO3 6

CO3 6

Q.5 Solve Any Two of the following.**12**

- A) Determine the Z- Transform and ROC of the following signals:
i) $x(n) = \{1, 2, 3, 4\}$ for $0 \leq n \leq 3$
ii) $x(n) = \{1, 2, 3, 4\}$ for $-1 \leq n \leq 2$
iii) $x(n) = \{1, 2, 3, 4\}$ for $-3 \leq n \leq 0$
iv) $x(n) = \delta(n)$
v) $x(n) = u(n)$
vi) $x(n) = a^n u(n)$
- B) Derive expression of $y(n)$ and draw the direct form- I and direct form- II realizations of the transfer function given by

CO3 6

$$H(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 + z^{-1} - z^{-2}}$$

- C) Explain the concept of an ideal filter. What are the key requirements for a filter to be considered ideal?

CO4 6

Q. 6 Solve Any Two of the following.**12**

- A) If $H(z)$ was designed using bilinear transformation, determine its parent causal analog transfer function

CO5 6

$$H(z) = \frac{4(z^2 + 3z + 4)}{10z^2 + 4z + 6}$$

- B) Derive the expression to obtain the order of a Butterworth filter.
- C) Explain the applications of multirate digital signal processing.

CO5 6

CO5 6

***** End *****

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech Branch : Instrumentation Engineering/ Instrumentation &Control Engg

Subject Code & Name: Digital Signal Processing, BTINC503

Semester :V

Max Marks: 60

Date:11/02/2025

Duration: 3 Hr.

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	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 The advantages of the butterfly structure is:	CO1	1
a. Reduces computation complexity.		
b. Requires a fewer number of multiplications and additions		
c. Combines the result of small DFTs into larger DFTs		
d. All of the above		
2 IDFT is converting.	CO1	1
a. frequency domain into time domain		
b. time domain into frequency domain		
c.		
d.		
3 DIT?-	CO1	1
a. Digital in time domain		
b. Discrete in time domain		
c. Decimation in time domain		
d. none of the above		
4 Wavelet series equation is the sum of	CO2	1
a. scaling coefficient		
b. detail coefficient		
c. span coefficient		
d. Both A and B		
5 One that is not a part of digital filter	CO2	1
a. unit delay		
b. multiplier		
c. delay		
d. subtraction		
6 Which of the following bus is used in the Digital signal processor?	CO2	1

	a. Program memory bus	b. Data memory bus	c. Both (a) and (b)	d. None of the above		
7	Which of the following form is used for the IIR filters?				CO3	1
	a. Direct form-I	b. Indirect form-I	c. Direct form-III	d. Direct form-IV		
8	CWT stands for				CO3	1
	a. Complete wavelet transform	b. Complex wavelet transform	c. Continuous wavelet transform	d. Cosine wavelet transformation		
9	What is the circular convolution of the sequences $x_1(n) = \{1, 1, 1, 1\}$ and $x_2(n) = \{1, 2, 3, 4\}$?				CO3	1
	a. $\{14, 14, 16, 16\}$	b. $\{16, 16, 14, 14\}$	c. $\{10, 10, 10, 10\}$	d. $\{14, 16, 14, 16\}$		
10	DIT algorithm divides the sequence into				CO1	1
	a. Positive and negative values	b. Upper higher and lower spectrum	c. Small and large samples	d. Even and odd samples		
11	DFT? -				CO2	1
	a. Discrete Fourier Transform	b. Digital Fourier Transform	c.	d.		
12	IDFT of the sequence $\{1, 0, 1, 0\}$ is				CO3	1
	a. $\{1, 0, 0, 1\}$	b. $\{0.5, 0, 0.5, 0\}$	c. $\{0.5, 1, 0.5, 0\}$	d. None of the above		
Q. 2 Solve the following.						12
A) Find the DFT of the following sequence: $x(n) = \{1, 1, 0, 0\}$					CO1	6
B) Obtain the circular convolution of the sequence: $x_1(n) = \{1, 2, 1\}$ & $x_2(n) = \{1, -2, 2\}$					CO1	6
Q.3 Solve the following.						12
A) Write a note on wavelet transform.					CO2	6
B) Compare STFT with FT.					CO2	6

Q. 4 Solve Any Two of the following.

12

- A) Realization of FIR filter using Direct form of given system:

CO3 6

$$y(n)=0.45 y(n-1)-0.2 y(n-2)+x(n)+2x(n-1)$$

- B) Explain realization of FIR filter using cascade form.

CO3 6

- C) Design an ideal low pass filter with frequency response

CO3 6

$$H_d(e^{j\omega})=1 \quad \text{for } \pi/3 < |\omega| < \pi$$

=0 otherwise, Find the value of $h(n)$ for $N=11$, Find $H(z)$.

Q.5 Solve Any Two of the following.

12

- A) What are the types of realization of IIR filter? Explain any one in details.

CO3 6

- B) For the analog transfer function $H(s)=2/(s+10)(s+52)$ determine $H(z)$

CO3 6

using BLT method. Assume $T=1$ sec.

- C) Explain impulse invariance method to design IIR filter.

CO2 6

Q. 6 Solve Any Two of the following.

12

- A) Draw the block diagram of DSP processor and explain each block in details.

CO1 6

- B) Explain applications of DSP in Biomedical Instrumentation

CO2 6

- C) Write a note on applications of digital signal processing.

CO1 6

*** End ***